

Length of surreal product

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Conjecture Every surreal number has a unique sign expansion, i.e. function $f: \alpha \rightarrow \{-, +\}$, where α is some ordinal. This α is the length of given sign expansion and also the birthday of the corresponding surreal number. Let us denote this length of s as $\ell(s)$.

It is easy to prove that

$$\ell(s+t) \leq \ell(s) + \ell(t)$$

What about

$$\ell(s \times t) \leq \ell(s) \times \ell(t)?$$

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